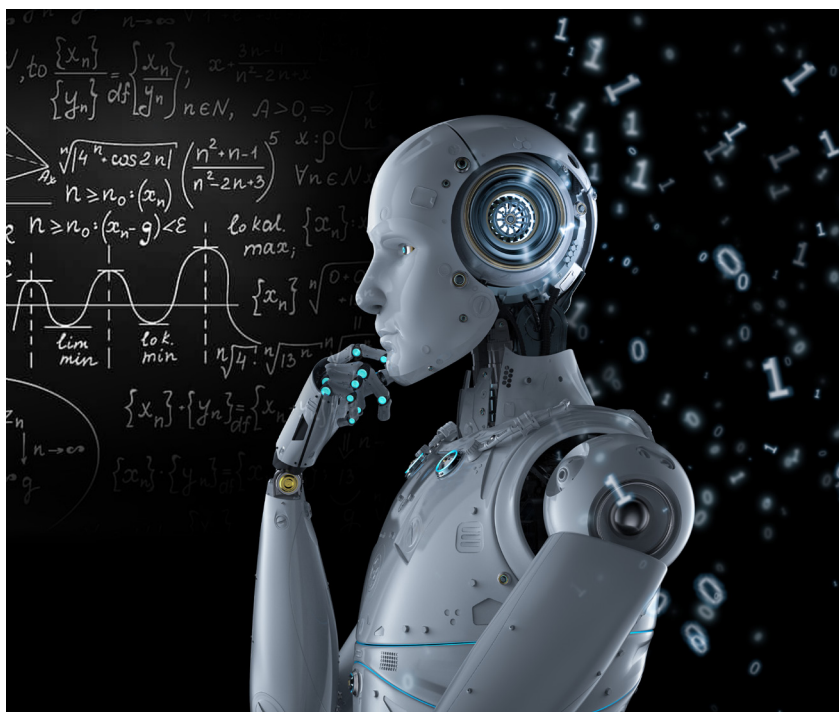


AI: Some History and a Lot More about Matrices

In the first article on this series on AI (published in *OSFY*, June 2022), we discussed how fields like AI, machine learning, deep learning, data science, etc, are related yet distinct from each other. We also made a few difficult choices regarding the programming language, tools, etc, which will be used throughout this series. Finally, we started discussing matrices. In this article, we will discuss more about matrix theory, the heart of AI. But before we proceed any further, let us discuss the history of AI.



people. The quest for creative machines outwitting human intelligence was also fascinating to us from ancient times. There were several failed attempts to make machines capable of playing chess. For example, Mechanical Turk was an infamous fake chess playing machine constructed in the late 18th century. The inventions of logarithm by John Napier, the Pascal's calculator by Blaise Pascal, difference engine by Charles Babbage, etc, were all predecessors to the development of artificial intelligence. Further, if you go through the history of mankind, you will find countless other real and imaginary (fake) moments hoping to achieve intelligence beyond the reach of the human brain. However, irrespective of all these achievements, the search for true AI began with the invention of digital computers.

However, if that is the case, the important question is: what were some of the milestones in the history of AI up until today? As mentioned earlier, the invention of digital computers is the single most significant event in the search for artificial intelligence. Unlike electro-mechanical devices where scalability was dependent on power requirements, digital devices simply benefited technological advancement like the transition from vacuum tubes to transistors to integrated circuits to present day very large-scale integration (VLSI) technology.

First of all, why is a discussion on the history of AI essential?

Well, there were a few times in history when there was great enthusiasm around research on AI, but on many of those occasions the huge expectations about the potential of AI went wrong. So, we need to learn some history of AI to understand whether this time we really are going to do wonders with AI or is it yet another bubble that is about to burst.

One important question that needs to be answered is: when did our fascination for artificial intelligence start? Was it after the invention of digital computers or did it start even before? I believe the quest for an omniscient being capable of answering all questions dates back to the beginning of civilisation. The Oracle of Delphi in the ancient times was one such supposedly divine being capable of answering all the queries asked by